Business Intelligence Internship Assignment

Insurance Policy & Claims Analytics

1. Objective

The objective of this assignment is to simulate a realistic insurance dataset and analyze policy sales and claims activity in order to evaluate portfolio profitability, claim patterns, and potential risk exposure. The analysis aims to help the insurance company understand how premiums compare with claims and identify key insights that can guide future pricing and risk strategies.

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2. Dataset Creation

Two datasets were created based on the assumptions provided in the assignment.

Policy Sales Dataset

The policy sales dataset simulates insurance policies purchased by customers in the year 2024. The dataset contains 1,000,000 records representing individual customers who purchased car insurance policies.

Each record includes the following fields:

- Customer_ID – Unique identifier for each customer
- Vehicle_ID – Unique identifier for each insured vehicle
- Vehicle_Value – Fixed value of ₹100,000 per vehicle
- Premium – Premium amount calculated as ₹100 multiplied by policy tenure
- Policy_Purchase_Date – Date when the vehicle and policy were purchased
- Policy_Start_Date – Policy activation date (Purchase Date + 365 days)
- Policy_End_Date – Date when policy coverage ends
- Policy_Tenure – Policy duration (1, 2, 3, or 4 years)

Policy Tenure Distribution

Policies were distributed according to the following proportions:

- 20% – 1 year policies
- 30% – 2 year policies
- 40% – 3 year policies
- 10% – 4 year policies

Policies were evenly distributed across all days in 2024 to ensure uniform policy sales throughout the year.

Claims Dataset

The claims dataset simulates insurance claims filed by customers after their policies become active.

Claim Rules for 2025

30% of vehicles purchased on the following dates file a claim:

- 7th
- 14th
- 21st
- 28th

Claims occur on the **first day of policy activation**.

No claims occur for vehicles purchased on other days.

Each vehicle can file **only one claim per year**.

Claim amount is calculated as:

Claim Amount = 10% × Vehicle Value
Claim Amount = ₹10,000

Claim Rules for 2026

Between **January 1 and February 28, 2026**, 10% of all vehicles with **4-year policy tenure** file a second claim.

These claims are evenly distributed across the 59 days within this period.

A vehicle that filed a claim in 2025 is eligible to file another claim in 2026 if it meets the eligibility criteria.

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3. Tools Used

The following tools were used to perform the analysis:

Python (Pandas) – Data generation and cleaning
SQL – Exploratory Data Analysis and aggregation queries
Excel – Initial validation and data verification
Power BI – Interactive dashboard creation and KPI visualization

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4. Key Performance Indicators (KPIs)

The following KPIs were calculated to evaluate insurance portfolio performance.

Total Policies

Total number of insurance policies sold.

Total Policies = 1,000,000

Total Premium Collected

Premium depends on policy tenure.

1 Year Policy → ₹100
2 Year Policy → ₹200
3 Year Policy → ₹300
4 Year Policy → ₹400

Total Premium Collected:

₹240,000,000
(₹24 Crore)

Total Claim Cost

Claims were calculated based on claim eligibility rules and claim amount.

Total Claim Cost (2025):

₹393,410,000

Total Claim Cost (2026):

₹100,000,000

Total Claim Cost:

₹493,410,000

Loss Ratio

Loss Ratio = Total Claims ÷ Total Premium

Loss Ratio = 493M ÷ 240M

Loss Ratio ≈ 205%

This indicates that the insurance portfolio is **currently unprofitable**, as claims exceed premium revenue.

Claim Frequency

Claim Frequency = Total Claims ÷ Total Policies

This metric measures the proportion of insured vehicles that file claims.

Average Claim Amount

Each claim equals 10% of vehicle value.

Average Claim Amount = ₹10,000

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5. Premium Earned Calculation

Insurance companies recognize premium gradually over the policy period.

Daily Premium was calculated as:

Daily Premium = Premium ÷ Total Policy Tenure Days

Premium earned was calculated from the policy start date up to **February 28, 2026**.

This represents the revenue already recognized by the company.

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6. Potential Future Claim Liability

Vehicles that have not yet filed claims may still file a claim during the remaining policy period.

Remaining Vehicles:

1,000,000 – Claims Filed

Potential Liability:

Remaining Vehicles × ₹10,000

Estimated Future Claim Liability:

Approximately ₹950 Crore

This represents the maximum potential claim exposure for the insurer.

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7. Dashboard & Visualization

An interactive Power BI dashboard was created to visualize policy performance and claims activity.

Dashboard Sections

Portfolio Overview

- Total Policies
- Total Premium
- Total Claims
- Total Claim Cost
- Loss Ratio

Claims Analysis

- Monthly Claim Trend
- Claim Cost by Year
- Claims by Policy Tenure

Profitability Analysis

- Premium vs Claims
- Loss Ratio by Policy Tenure
- Future Claim Liability

Interactive slicers allow filtering by:

- Policy Tenure
- Claim Year
- Claim Type

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8. Key Business Insights

1. **Loss Ratio exceeds 200%**, indicating claims are significantly higher than premium revenue.
2. Claims increase sharply in **2025**, as policies begin after the 365-day waiting period.
3. **3-year policies generate the highest premium revenue**, as they represent 40% of the portfolio.
4. Long tenure policies increase the **risk exposure window**, increasing the probability of claims.
5. Claim occurrence patterns based on purchase dates indicate **risk clustering**.

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9. Conclusion

The simulated analysis highlights the importance of balancing premium pricing and claim exposure in insurance portfolios. While the portfolio generates significant premium revenue, the high claim cost results in an unfavorable loss ratio.

To improve profitability, insurers may consider:

- Adjusting premium pricing
- Reducing claim frequency through risk assessment
- Diversifying policy tenure distribution

- Implementing stricter underwriting rules

The Power BI dashboard provides an effective way for stakeholders to monitor these metrics and make informed business decisions.

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